



	Blood Vessel Involved	Leads to be Checked
Lateral Wall MI	LCX	• I, aVL, V5, V6
Septal MI	Septal branch of LAD	• I, aVL, V1, V2
Anterior Wall MI	Distal LAD	• I, aVL, V2-V4
Extensive Anterior wall MI	Proximal LAD	• I, aVL, V1-V6
Inferior Wall MI	RCA	• II, III, aVF
Posterior Wall MI	Post-descending artery	• V2/V3: Horizontal ST ↓, upright T wave • For confirmation: V7, V8, V9
Anterolateral Wall MI	LAD, LCX	• I, aVL, V3-V6

INFERIOR WALL MI

00:47:54

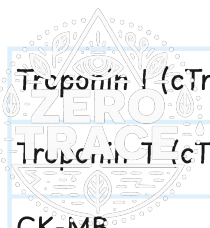
- Blood vessel involved: RCA [**Right Ventricle**]
- Nerve that gets involved:
 - Vagus irritation → Bradycardia, Vomiting (reverse peristalsis)
 - Phrenic nerve irritation → Hiccoughs
- ECG findings seen in: Leads II, III, aVF
- Vitals status: HR ↓, BP ↓, JVP ↑
 - Bilateral crepitations absent (no pulmonary edema)
- Drugs contraindicated:
 - Metoprolol
 - S/L NTG
- Drugs indicated:
 - Atropine
 - IV fluids → to ↑ BP > 90/60 mmHg

CARDIAC BIOMARKERS

00:53:29

- **IOC for MI: Elevated cardiac biomarkers**
 - Highest sensitivity: Troponin I (cTnI) and Troponin T (cTnT)

Protein	First Detection	Sensitivity	Specificity
Ischemia Modified Albumin	< 1 hour	+++	+
Heart Fatty Acid Binding Protein	1.5-2 hours	+++	++
Myoglobin	1.5-2 hours	+++	+



Troponin I (cTnI)	3-4 hours	++++	++++
Troponin T (cTnT)	3-4 hours	++++	++++
CK-MB	4-6 hours	++	++
LDH	6-10 hours	++	+

- **Myoglobin** is one of the earliest markers to rise
 - But it can also rise d/t skeletal muscle damage

Biomarker	Onset of Initial Rise	Peak Serum Level	Return to Baseline
CK-MB	4-6 hrs	24 hrs	48-72 hrs
cTnI	2-4 hrs	24-48 hrs	7-10 days
cTnT	2-4 hrs	24-48 hrs	10-14 days
H-FABP	1.5 hrs	4-6 hrs	20 hrs
IMA	Minutes	6 hrs	12 hrs
Myoglobin	1-4 hrs	6-12 hrs	24 hrs

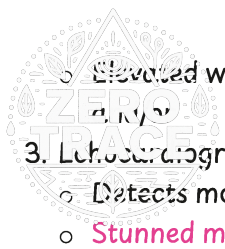
- STEMI progresses through the following stages
 1. Acute stage → Hours to first 7 days
 2. Healing stage → 7 to 28 days
 3. Healed stage → >29 days
- < 1 hour: **Golden period of MI** → Highest chances of Arrhythmia (V-Fib, PVT)
- Earliest cardiac biomarkers to rise in myocardial infarction:
 1. Ischemia Modified Albumin
 2. Heart Fatty Acid Binding Protein
 3. Myoglobin
- Most Sensitive and Specific Cardiac Biomarker for Diagnosis of MI: **Troponins (cTnI, cTnT)**
 - Rise by 4 hours after onset of MI
- **Early** Reinfarction in MI: **CPK-MB**
- Reinfarction in MI: **Trop I** (↑ > 20% over initial value)

INVESTIGATIONS DONE IN ANY CASE OF CHEST PAIN AT REST / SUSPECTED ACUTE CORONARY SYNDROME

01:02:57

CHEST PAIN THAT RADIATES TO ARM, SHOULDER, NECK, JAW, WITH DIAPHORESIS

1. ECG: STEMI or NSTEMI
and/or
2. Cardiac Biomarkers: IOC



- Elevated with rise and fall, with one value > 99th percentile of URL
- 3. Echocardiography:
 - Detects motion wall abnormality, rules out Takotsubo cardiomyopathy
 - **Stunned myocardium**: Sudden decrease in contractility of the heart and/or
- 4. Coronary Angiography: Detects thrombus causing complete occlusion and/or
- 5. Autopsy

MANAGEMENT OF STEMI

01:08:56

- Treatment of Choice: Primary PCI / Balloon Angioplasty
- Patient came to a PCI-capable hospital:
 - Send to Cath lab for primary **PCI** when FMC (First Medical Contact) to device time < **90 minutes**
- Patient came to a non-PCI-capable hospital:
 - Start **fibrinolysis** within **30 minutes** if anticipated FMC to device time > **120 minutes**
 - Transfer for PCI if FMC to device time is < 120 minutes

DRUG TREATMENT IN STEMI

Mnemonic: **BEST-MOAN**

1. DAPT (DUAL ANTIPLATELET THERAPY):	• Aspirin (TxA2 inhibitor) 325 mg chewable + Ticagrelor (P₂Y₁₂ inhibitor)
2. SpO2 MONITOR	• < 93% on room air → use Hudson mask @ 6 L/min
3. NITRATES	• S/L NTG 0.4 mg, repeat @ 5 min, max = 3 doses • Contraindication: SBP < 90, Inferior wall MI
4. MORPHINE	• Vagotonic action → ↓ HR → ↓ O₂ consumption of heart
5. METOPROLOL	• Single dose of beta-blocker has shown substantial reduction in mortality as it ↓ HR, ↓ O₂ consumption, ↓ infarct size • Contraindications: ◦ Acute HF, ◦ Bradycardia, ◦ Inferior wall MI
6. ENOXAPARIN	• Via subcutaneous route
7. ACEI AND STATINS	• ACEI: for HTN • Statins: For plaque stabilization



THROMBOLYSIS IN NON-PCI EQUIPPED HOSPITALS

- **Door to Needle Time: < 30 minutes**
- Side effect of fibrinolysis: Bleeding
- Upper limit of fibrinolysis: ~12 hours of symptom onset

DRUGS USED

1. Alteplase
2. Reteplase → Bolus fibrinolytic
3. Tenecteplase → Bolus fibrinolytic
4. Streptokinase (STK)
 - Obtained from Group C Streptococcus (Streptococcus Equisimilis)
 - Anaphylaxis risk +

CLEAR CONTRAINDICATIONS TO THROMBOLYTICS

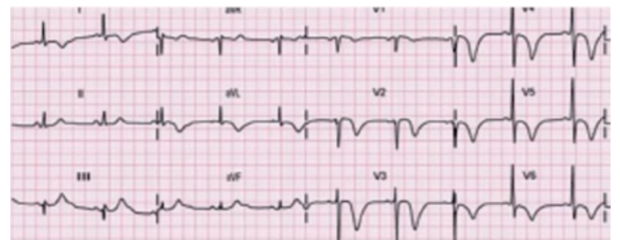
(Mnemonic: HAS-BLEED)

1. **H**: Hypertension (BP > 180/110 mmHg) at any time during presentation
2. **A**: Aortic dissection suspected (Coronary Artery dissection)
3. **S**: Stroke (hemorrhagic at any time)
4. **S**: Stroke (non-hemorrhagic in last 1 year)
5. **B**: Bleeding (active internal bleed, excluding menstruation)

NSTE-ACS (NSTEMI/Unstable angina)

01:23:52

- NON-ST elevation- Acute coronary syndrome → Spectrum
- Severe chest discomfort with any one of three:
 1. Occurs at rest/minimal exertion and lasting >10 mins
 2. Recent onset (last 2 weeks)
 3. Crescendo pattern (↑ severity, ↑ duration, ↑ frequency)
- Radiation of Pain: Jaw, arm, shoulder
- ECG: ST depression and/or T-wave inversion
- Biomarkers: > 99th percentile of URL



Important Information

- Differentials in acute coronary syndrome:
 1. UA: ST ↓, T wave inversion (biomarkers are normal)
 2. NSTEMI: ST ↓, T wave inversion
 3. STEMI: ST ↑, T wave inversion (elevated biomarkers)

TIMI RISK SCORE FOR NSTE-ACS

01:29:06

- Parameters
 1. Age > 65 years
 2. Known CAD with > 50% stenosis

3. ST depression of > 0.5 mm
 4. ↑ Cardiac biomarkers
 5. 1-2 episodes in last 24 hrs
 6. Prior angina
 7. > 3 CAD risk factors
- If score > 7: **41%** risk of adverse cardiac event

MANAGEMENT

01:30:51

Mnemonic: BEST-MOAN

- **B** - B-blocker
 - **E** - Enoxaparin
 - **S** - Statin
 - **T**
 - Thrombolysis: **CI** in **NSTEMI and UA**
 - **Tirofiban**: IV antiplatelet drug used in NSTEMI/UA
 - **M** - Morphine
 - **O** - Oxygen
 - **A** - Aspirin
 - **N** - Nitroglycerin
- } Aspirin → Oxygen → S/L NTG → Morphine

ANTI-ISCHEMIC DRUGS	• N: NTG SL • B: Beta-blockers (↓ HR) • M: Morphine (vagolytic) • Consider CCB if β-blockers are contraindicated
ANTI-THROMBOTIC	• Antiplatelets (DAPT): Aspirin + Ticagrelor • IV Antiplatelets before PCI: ◦ Drugs: Tirofiban, Cangrelor, Eptifibatide (gp IIb/IIIa inhibitors) ◦ Very high risk → PCI within 2 hrs ◦ High risk: PCI within 24 hours • Anticoagulant: Enoxaparin

TYPES OF MYOCARDIAL INFARCTION

01:39:40

Type 1	• Coronary artery thrombus • Dissection
Type 2	• Operative stress • Anemia (oxy-Hb ↓) • Arrhythmia • Heart failure • Valvular disease • Cocaine overdose (vasoconstrictor → ischemia)



Type 3	Sudden cardiac death
Type 4A	During PCI and cTn values >5 times URL
Type 4B	Due to Stent thrombosis
Type 4C	Due to restenosis
Type 5	During CABG and cTn values >10 times URL

COMPLICATIONS OF MYOCARDIAL INFARCTION

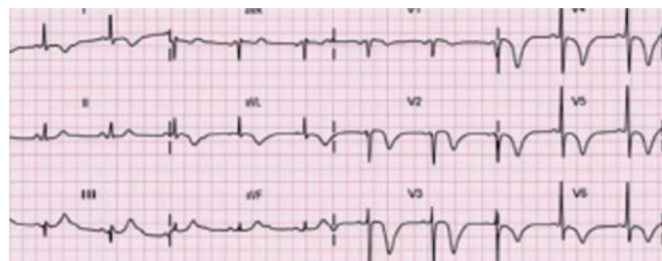
01:45:28

A. Arrhythmias	- Ventricular Fibrillation, Pulseless Ventricular Tachycardia - Cause of sudden cardiac death (SCD)
B. Brady-arrhythmias	- Often due to Mobitz II heart block - Cause of sudden cardiac death (SCD)
C. Cardiogenic Shock	- Mortality occurring after 24 hours of admission - Major contributor to in-hospital mortality
D. Dressler Syndrome	- Occurs >6 weeks post-MI - Etiology: Autoimmune pericarditis/pleuritis - Features: → Chest pain at rest → Pericardial friction rub → ST elevation (concave upwards, global) - Treatment: Aspirin, avoid steroids
E. Valvular Lesions	M/c valvular lesion seen after MI is Mitral regurgitation

Q. A 55-year-old male with HTN and hyperlipidemia presents to the clinic due to chest pain for 6 hours. ECG is shown. Which of the following is not to be used in this patient?

- Alteplase
- Aspirin
- Morphine
- Metoprolol

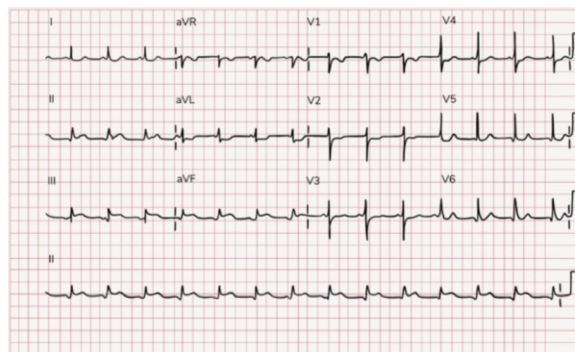
Ans: (a)



Q. A diabetic patient presents with chest pain for the last 4 hours with uneasiness. BP= 150/110 mm Hg and early systolic murmur is heard. ECG is done. Correct diagnosis?

- Anterior wall MI
- Posterior wall MI
- Inferior wall MI
- Lateral wall MI

Ans: (b)



Q. A 50-year-old man presents with a history of recurrent retrosternal chest pain episodes at rest lasting about 5 minutes and partially subsiding with sublingual nitrate. ECG shows left ventricular hypertrophy and flat T wave. He is a known case of HTN, DM, and hypercholesterolemia currently on aspirin, atenolol, metformin, and lovastatin. What is the next big step in management?

- IV glyceryl trinitrate infusion
- Injection enoxaparin
- Add clopidogrel
- Increase the dose of beta blocker

Ans: (b)



16. SYMPTOMATIC BRADYCARDIA WITH A PULSE

INTRODUCTION

00:00:41

- Marathon runner:
 - Asymptomatic bradycardia
 - Physiological, d/t high vagal tone
- Hypothyroidism:
 - $\downarrow T4, \downarrow T3 \rightarrow \downarrow$ sympathomimetic stimulation $\rightarrow \downarrow$ HR
 - Systolic BP Normal, Diastolic BP $\uparrow$
 - No T/t required for Bradycardia; T/t done only for HTN
- Bradycardia: HR < 60 bpm
 - $BP = HR \times SV \times P. \text{ resistance}$
 - If HR $\downarrow \rightarrow$ Bp $\downarrow$



CHEMICAL PACING

DRUGS

- Atropine
- Epinephrine
- Dopamine

CLINICAL FEATURES

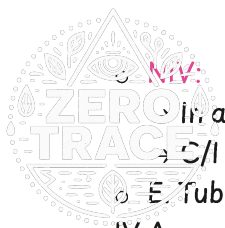
- Dizziness, Altered mentation
 - Bradycardia ($\downarrow$ HR) $\rightarrow \downarrow$ BP $\rightarrow \downarrow$ cerebral blood flow
- Chest pain:
 1. Inferior wall MI caused by the following (only MI with Bradycardia): can have
 - $\rightarrow$ Vomiting (d/t vagus nerve irritation)
 - $\rightarrow$ Hiccups (d/t phrenic nerve irritation)
 2. $\downarrow$ Coronary blood flow
- Dyspnea: D/t pooling of blood in lungs

ECG CHANGE

- R-R interval: > 5 large squares
- P-R interval: $\uparrow$
- R-R and P-R interval on ECG is inversely related to HR

TREATMENT

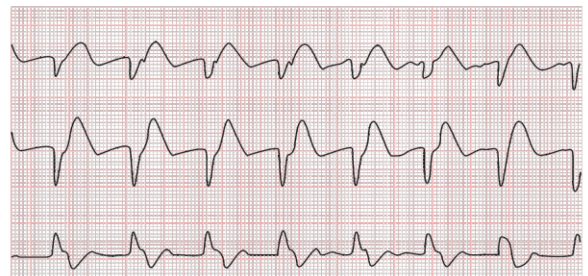
- Patency airway $\rightarrow$ Guedel airway
- SpO₂ Monitoring: $\downarrow$
- Assisted ventilation: NIV/ E. Tube



- In acute pulmonary edema
 - C/I in unconscious patients
 - E tube: In acute pulmonary edema, if patient is unconscious
- IV Access:
 - Atropine: 1 mg, max = 3 mg
 - 3 doses @ 3-5 min gap
 - If Atropine fails (Worsening of hemodynamic status) → **TCP** (Transcutaneous Pacer)
 - Worsening of hemodynamic status despite TCP → **TVP** tvp done when tcp fails (Trans Venous pacing)

TRANSCUTANEOUS PACING

- Non-invasive emergency procedure delivering electrical impulses via chest muscles to stimulate the heart
- Paddle placement:
 - 2nd ICS, right parasternal line
 - 6th ICS, left mid-axillary line
- Current is increased in 10 mA increments
- Effective pacing usually achieved by 70 mA
- On ECG:
 - Pacer spikes
 - Broad, slurred R wave and Broad S wave
- In patients with TCP done, there will be a fluttering sensation in the chest
- To check adequacy of perfusion:
 - Normally: carotid artery is used
 - In TCP: **Femoral Artery (as carotid gives false results)**



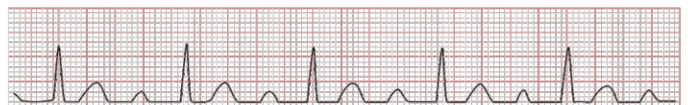
TRANSVENOUS PACING

- Used when there is worsening of hemodynamic status despite TCP

FIRST DEGREE HEART BLOCK

00:22:14

- Normal PR interval: 120-200 msec
- In 1st degree Heart block: PR > 210 msec
- Caused by delayed conduction between the SA and AV nodes
- In countries like India, Aschoff nodules due to rheumatic fever may cause conduction delay between SA and AV nodes
- ECG finding:
 - Prolonged PR interval
 - No missed beats
- Cause:
 - Physiological cause (High Vagal Tone): **MCC of 1st-degree Heart Block**
 - Aschoff nodule
 - Non caseating granulomas (sarcoidosis)
 - Iron deposits in hemochromatosis



SECOND DEGREE HEART BLOCK

00:26:01



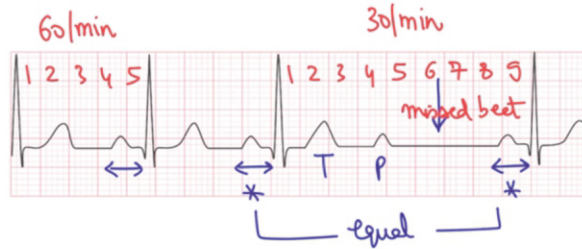
MOBITZ-I HEART BLOCK

- Aka Wenckebach Phenomenon



- ECG Findings
 - Serial $\uparrow$ in P-R interval
 - **Missed Beat:** Dropped QRS $\rightarrow$ $\uparrow$ R-R interval
 - **Marked asymmetry between P-R interval** before and after missed beat
 - $\rightarrow$ P-R before missed beat $\rightarrow$ longer
 - $\rightarrow$ P-R after missed beat $\rightarrow$ Shorter
- Beta blocker toxicity can contribute to Mobitz-I HB

MOBITZ-II HEART BLOCK

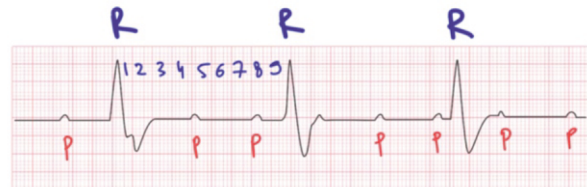


- ECG Findings
 - P-R interval is constant
 - Sudden drop in QRS (missed beat)
 - P-R interval prior and after missed beat are equal

THIRD DEGREE HEART BLOCK

00:32:39

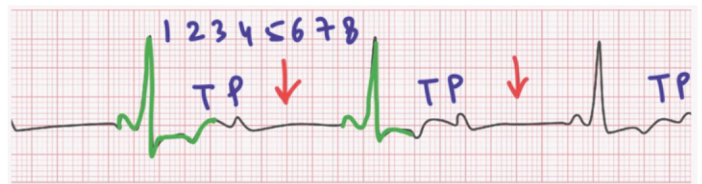
- ECG Findings
 - Number of P waves and number of R-waves do not match
- Aka Complete Heart Block / Stokes Adam syndrome
- A-V dissociation: **Cannon a waves** -(frog sign)



2:1 HEART BLOCK

00:36:09

- ECG Findings
 - Every alternate P wave is not followed by a QRS (dropped beat)
 - Appears as one conducted P wave followed by one blocked P wave





Q. Identify the conduction abnormality from the ECG given below?

- a. 1st degree heart block
- b. 2: 1 block
- c. 2nd degree heart block
- d. 3rd degree heart block

Ans: (c)

Q. A 60-year-old male patient presents to the clinic with complaints of occasional light headedness and dizziness. An electrocardiogram reveals a pattern consistent with Mobitz Type-I (Wenckebach phenomenon). Which of the following statements best describes this condition?

- a. Accelerated AV nodal conduction
- b. Progressive decrease in PR interval with each beat and a missed beat
- c. Progressive increase in PR interval with each beat and a missed beat
- d. The PR interval before and after the missed beat are equal

Ans: (c)



CONGENITAL HEART DISEASE



CONGENITAL HEART DISEASE

NADAS CRITERIA

00:01:02

MAJOR	MINOR
- Central Cyanosis (Right to left shunt) - Pulmonary oligemia - Non-responsive to O₂ administration - Congestive Heart Failure (Left to right shunt) - Pulmonary plethora - Diastolic murmur - Systolic murmur $\geq$ grade III	- Systolic murmur < Grade III - Abnormal S₂ - Single S₂ (Tetralogy of Fallot) - Abnormal ECG - Right axis deviation - Abnormal Chest X-ray - Boot-shaped heart - Abnormal BP values - Coarctation of aorta - Upper limb BP - $\uparrow$ - Lower limb BP - $\downarrow$
Presence of 1 major / 2 minor criteria suggest presence of congenital heart disease	

GRADING OF MURMURS (Subjective)

GRADE 1	Faint murmur identified by superspecialist (DM cardiologist)
GRADE 2	Can be identified by MD specialist
GRADE 3	Can be heard by Intern
GRADE 4	Can be felt by patient/patient attendant
GRADE 5	Loud murmurs that can be heard on placing stethoscope off 2 cm chest

Important Information

- Pathological Murmur suspected -
 - IOC - Transthoracic echocardiogram

ACYANOTIC CONGENITAL HEART DISEASES

00:11:04

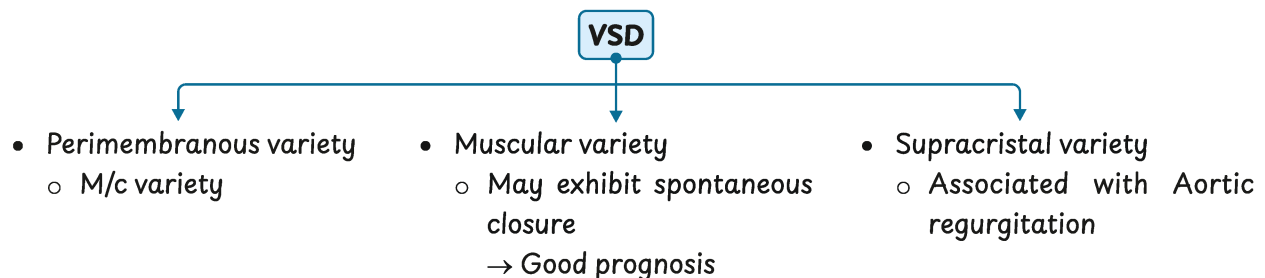
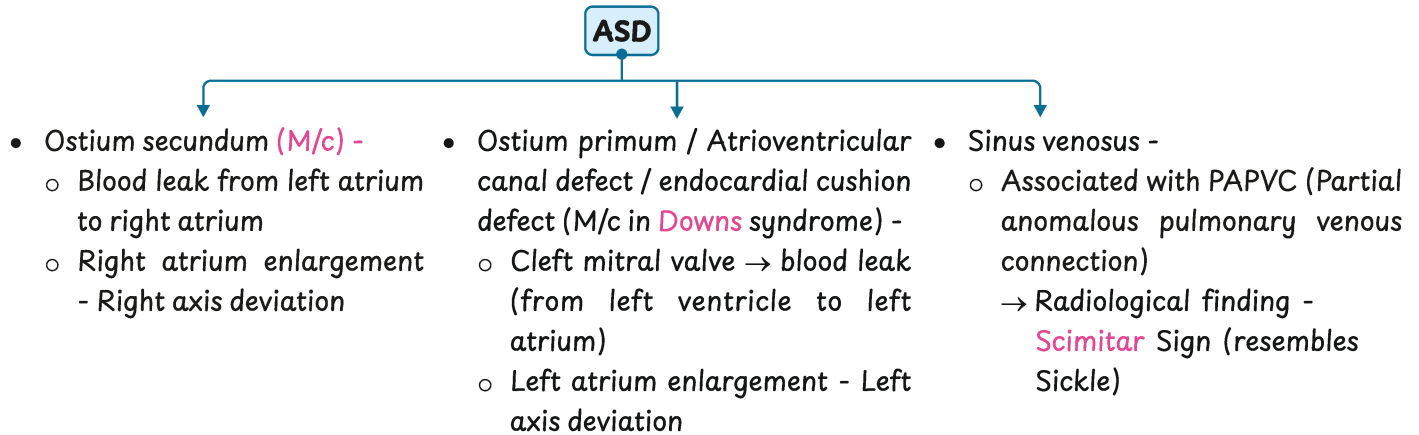
	VENTRICULAR SEPTAL DEFECT (VSD) (M/c)	PATENT DUCTUS ARTERIOSUS (PDA) Pressure difference (Aorta-Pulmonary Artery) = $\uparrow$ pressure shunt	ATRIAL SEPTAL DEFECT (ASD) Pressure difference (Left Atrium-Right Atrium) = 4 mmHg ($\downarrow$ pressure shunt)
C/F	Congestive heart failure + Recurrent pneumonia episodes		
AGE OF PRESENTATION	6 weeks	Preterm child- at birth- Birth, term child- at 6 weeks	5 years, Delayed onset - Adult



	Wide split (variable)	Narrow split	Wide split (fixed)
	Pan systolic murmur	Continuous murmur	- Shunt murmur - Absent - Flow murmur (Mid Diastolic Murmur/ Ejection systolic Murmur) - present
INTERVENTION	Dacron patch	- Preterm - Indomethacin - Term- Surgical ligation	Occlusion devices used

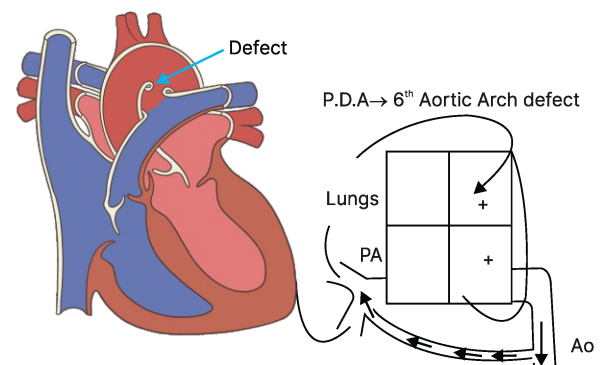
ATRIAL SEPTAL DEFECT & VENTRICULAR SEPTAL DEFECT

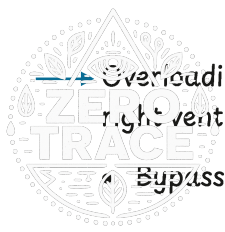
00:20:45



PATENT DUCTUS ARTERIOSUS

- 6th aortic arch defect
 - Left-to-right shunt
- Connection between aorta and pulmonary artery
- Ductus arteriosus
 - Begins from bifurcation of pulmonary artery
 - Ends at starting of descending aorta
 - Closure -
 - Begins to close - 15 hours after birth (normally)
 - Functional closure - 7th day
 - Anatomical closure - 28th day
- Blood in aorta shunted to lungs → Overloading of lung circulation → pulmonary edema





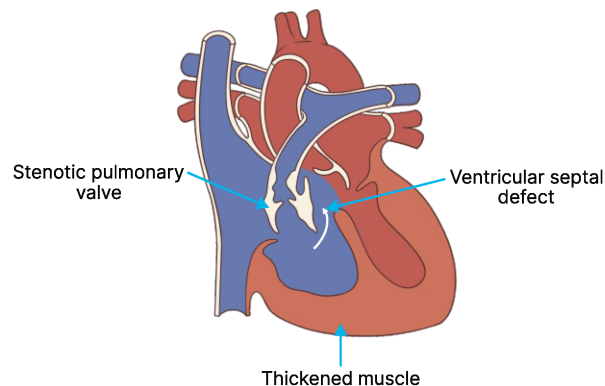
- Overloading of lung circulation → blood rushes to left side of heart → left ventricular failure (initial) → right ventricular failure (later)
- ↳ Bypasses right ventricle initially

CYANOTIC HEART DISEASES

00:30:40

TETRALOGY OF FALLOT

- M/c congenital heart disease in infancy and beyond



TETRALOGY OF FALLOT

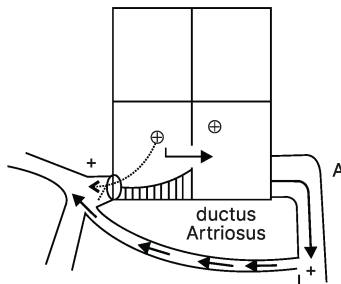
- Right Ventricular Outflow Tract narrowing - Subpulmonic stenosis
 - **Hallmark feature**
 - Major determinant of central cyanosis
 - Causes Pulmonary Oligemia
- Concentric Right Ventricular Hypertrophy
 - Pressure change (Right ventricle > Left ventricle)
- Right to Left shunt - Overriding aorta - Ventricular septal defect
 - Receives mixed blood
 - Mixing of 2 circulations
 - Development of Central cyanosis
 - Non-responsive to O₂ administration

TRILOGY OF FALLOT	PENTALOGY OF FALLOT
• Right Ventricular Outflow Tract narrowing - Subpulmonic stenosis • Concentric Right Ventricular Hypertrophy • Atrial Septal Defect (ASD)	• Tetralogy of Fallot + ASD

WORK UP OF TETRALOGY OF FALLOT

ON EXAMINATION	CHEST X-RAY	IOC	T/t
• Cyanosis (non-responsive to oxygen) • Digital clubbing • S₂ - A₂ P₂	• Concentric Right Ventricular hypertrophy ◦ Boot shaped heart (Coeur en Sabot)	• Transthoracic Echo (TTE)	• DOC - Alprostadil ◦ Administered before functional closure of ductus arteriosus

○ Single S₂ heard → P₂ is poorly heard • Ejection systolic murmur /Crescendo-Decrescendo murmur - ○ Intensity of murmur becomes softer during Tet spell	• CT ratio - Normal • Upturned apex • Pulmonary oligemia • Right-sided aortic arch 	○ Patency of ductus arteriosus maintained → Ductal dependent pulmonary circulation maintained • Palliative surgery - Blalock Taussig shunt ○ Connects subclavian artery with pulmonary artery ○ ↑ pulmonary blood flow
---	---	--



TET SPELLS / HYPER CYANOTIC SPELLS

- Seen in Tetralogy of Fallot
- Central cyanosis present - Lips - becomes blue → black
- ↑ Shunting from Right ventricle to left ventricle → ↓ blood supply to lungs → worsening of cyanosis
 - Ejection systolic murmur becomes softer
- Management of Tet spells
 - **Knee-chest position**
 - Causes kinking of femoral artery → ↑peripheral resistance→↑left ventricle pressure → Right to left shunting ↓ → Improves cyanosis
 - I/V Morphine - for sedation
 - I/V propranolol - to control HR, peripheral resistance
 - I/V Methoxamine
 - ↑ Systolic BP → ↑left ventricular pressure (to overcome resistance) → Right to left shunting ↓ → Improves cyanosis
 - I/V Soda bicarbonate - to neutralize acidosis



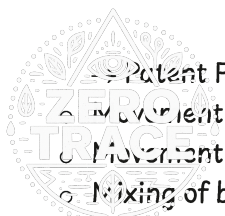
Important Information

- I/V calcium gluconate → **No role** in management of Tet spells

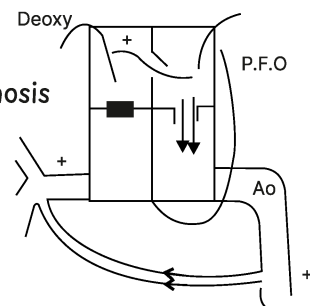
TRICUSPID ATRESIA

- Pathophysiology
 - Absence of tricuspid valve (Since birth)
 - Right atria pressure > Left atria pressure

01:00:10



- ⇒ Patent Fossa Ovalis occurs
- Movement of deoxygenated blood from Right atria to left atria
- Movement of oxygenated blood from lungs to left atria
- Mixing of blood in left atria → pumped into aorta → development of central cyanosis
- Central cyanosis on **day 0** of birth
- Left-sided enlargement of heart
- Blood supply of lungs - Only via Ductus Arteriosus (mixed blood)
 - Ductal-dependent pulmonic circulation
 - Ductus arteriosus facilitates survival of patient
 - If Ductus arteriosus functionally closes on day 7 → Death on day 7 (after birth)
- Drug administered to maintain ductal patency (after birth) - Alprostadil



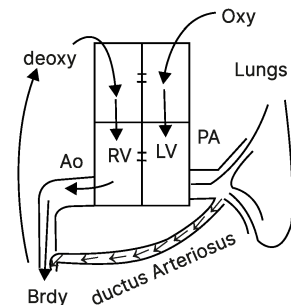
FINDINGS

- Chest X-ray - **CT ratio** ↑ (Left-sided enlargement)
- Auscultatory findings
 - S₁ - ↑ Intensity M1 → Single S₁
 - T₁ absent
 - S₂ - A₂ is normal → Single S₂ (No T₂)

TGA (TRANSPOSITION OF GREAT ARTERIES)

01:07:50

- Aorta originates from RV & pulmonary artery originates from LV - 2 parallel circulation
 - Deoxygenated blood enters right side of heart (RA → RV) → pumped back into body via aorta → returns to RA
 - Oxygenated blood (From lungs) pumped into left side of heart (LA → LV) → pumped back into body via pulmonary artery → returns to LA
- Mixing of blood occurs by ASD/VSD/Ductus arteriosus
- Oxygenated blood reaches systemic circulation only via ductus arteriosus
 - Ductus arteriosus facilitates survival of patient
 - If Ductus arteriosus functionally closes → Death
- Central cyanosis on **day 0** of birth



T/t

Main surgery- **Switch** operation
(Cardio-pulmonary bypass)

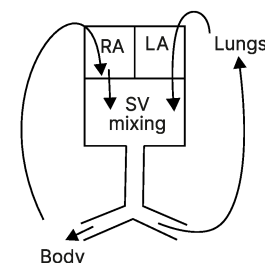
Drug administered to maintain ductal patency (after birth)- Alprostadil

- Chest X-ray - **Egg on side** appearance

TRUNCUS ARTERIOSUS / CO-TRUNCAL ABNORMALITY

01:13:55

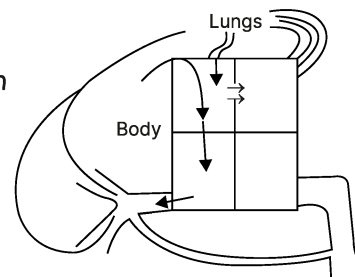
- Single ventricle present
 - Mixing of oxygenated & deoxygenated blood in single ventricle
 - Mixed blood pumped into body → returns back (to right atrium)
 - Mixed blood pumped into lungs → returns back (to left atrium)
- Development of Central cyanosis
- **Alprostadil** → C/I
 - No presence of aorta & pulmonary artery
- **Bad** prognosis



TOTAL ANOMALOUS PULMONARY VENOUS CONNECTION (TAPVC)

01:17:20

- Normal circulation - Deoxygenated blood pumped into right side of heart → pumped to pulmonary artery → reaches lungs → Pulmonary veins arise from lungs (pump oxygenated blood)
- Normal pulmonary veins not present
 - Abnormal blood vessels (arising from each lung) → pumps oxygenated blood to right atria → Overloading of right-side circulation → mixing of blood occurs → cyanosis
- Associated Atrial Septal Defect
 - Ensures mixed blood reaches left side of heart → reaches systemic circulation
 - Wide fixed split S2
- **Alprostadil → C/I**
 - Overloading of overloaded RA



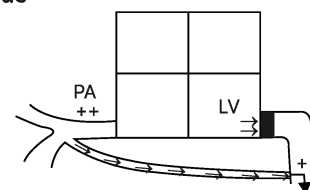
TYPES BASED ON ABNORMAL CONNECTION ROUTE

- SVC - Supracardiac type
 - **Figure of 8 / snowman heart** appearance on Chest X-ray
- RA - Cardiac type
- IVC - Infracardiac type

AORTIC ATRESIA

01:23:00

- Left ventricle cannot pump blood to aorta
 - Cause - Aortic valve defect
- Pulmonary artery pressure > Aorta pressure
- Deoxygenated blood from pulmonary artery pumped into aorta via ductus arteriosus
 - Ensures survival
- Neonate develops
 - Shock → negligible pulses
 - **Central cyanosis**



T/t

Alprostadil - Maintains Ductal patency
(↑ survival of infant by few hours)

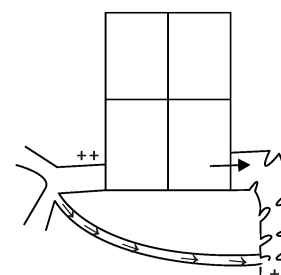
Heart Transplantation from
age-matched donor

- Ductal-dependent systemic circulation

INTERRUPTED AORTIC ARCH

01:26:00

- Presence of Aortic valve → **Tedious** blood flow to aorta
- Pulmonary artery pressure > Aorta pressure
- Deoxygenated blood from pulmonary artery pumped into aorta via ductus arteriosus
 - Ensures survival
- T/t - Alprostadil - Maintains Ductal patency
- Ductal-dependent systemic circulation



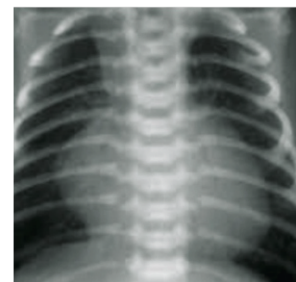
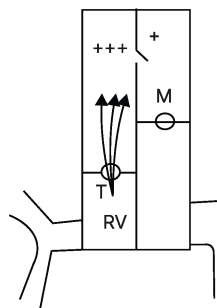


DUCTAL DEPENDENT CIRCULATION		DUCTAL INDEPENDENT CIRCULATION
PULMONARY	SYSTEMIC	
- Tetralogy of Fallot - Tricuspid atresia	- Aortic atresia - Interrupted aortic arch - Hypoplastic left heart syndrome - Mitral stenosis	- Kawasaki disease - Total Anomalous Pulmonary Venous Connection - Truncus arteriosus

EBSTEIN ANOMALY

01:30:16

- Cause - Maternal intake of lithium**
- Inferior displacement of malformed tricuspid valve
 - Mitral valve - normal position
- Size and pressure in RA > LA
 - Patent Fossa Ovalis** → mixing of blood → Cyanosis
- RV is smaller
 - ↓ blood pumped to lungs → Cyanosis
- Tricuspid regurgitation
 - Contraction of RV → leakage of blood to RA → ↑ pressure in RA
 - Pan systolic** murmur
- Association - Wolff-Parkinson-White (WPW) syndrome
- CXR - **Box-shaped** heart



Important Information

- Egg shell calcification - Sarcoidosis, Silicosis
- Egg in a cup appearance - Constrictive pericarditis

LEFT → RIGHT SHUNT (CHF + RECURRENT PNEUMONIA)	RIGHT → LEFT SHUNT (CYANOSIS, FAILURE TO THRIVE) SINGLE S ₂	MIXED DEFECTS (CYANOSIS + CHF)
- Ventral Septal Defect (wide variable S₂) - Atrial Septal Defect (wide fixed S₂) - Patent Ductus Arteriosus (Narrow split)	- Tetralogy Of Fallot - Tricuspid atresia - Ebstein anomaly	- Transposition of Great Arteries (Egg on side appearance) - Total Anomalous Pulmonary Venous Connection (Figure of 8 appearance)



Q. Which among the following is a major NADAS criteria?

- a. Abnormal S2
- b. Abnormal ECG
- c. Diastolic murmur
- d. Systolic murmur grade 1

Ans: (c)

Q. A 10-year-old boy is admitted to the hospital for pneumonia. Further evaluations reveal an atrial septal defect. There is no family history of heart disease except for ischemic heart disease in the grandfather. Which of the following is incorrect regarding atrial septal defect?

- a. Pulmonary plethora
- b. Wide fixed split S2
- c. Mid-diastolic flow murmur
- d. Ejection systolic shunt murmur

Ans: (d)

18. METABOLIC-SYNDROME-X AND SYNDROME-Z

METABOLIC SYNDROME / SYNDROME X FEATURES (MNEMONIC - CODE HEART)

00:00:30

00:00:31

DIAGNOSTIC FEATURES OF METABOLIC SYNDROME/SYNDROME X

Waist circumference -

Centripetal obesity

(Important aspect)

- American males >102 cm
- American Females >88 cm
- Indian male >90 cm
- Indian females >80 cm



Excess weight in abdomen

On specific DM medication/FBS >100-125 mg/dl (impaired fasting glucose)



High blood pressure

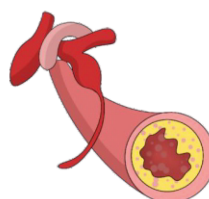
Hypertensive on medication/BP >130/85 mmHg

- ↑ in peripheral resistance



Elevated blood sugar levels

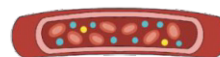
↑ Triglycerides >150 mg/dl



Hypertriglyceridemia

↓ HDL

- Men - <40 mg/dl
- Female - <50 mg/dl



Low levels of HDL cholesterol

- 3 / 5 features → Diagnosis for Metabolic syndrome / Syndrome X

Important Information

- Features of Insulin resistance, Acanthosis nigricans & Non Alcoholic Fatty Liver Disease → Can be seen in these patients
 - M/c cause of Cirrhosis -
 - M/c cause of development of Chronic liver disease -
 - M/c cause for Liver transplantation -

} Non-Alcoholic Fatty Liver Disease